MSS (2004) Revisited:
Weak Instruments and Dynamic Panel Identification

Alexander Lievens and Tobias Vanoverschelde
KU Leuven: D0S91a Advanced Applied Econometrics
Prof. Dr. Dirk Czarnitzki

May 2026

Abstract

We re-examine the rainfall-instrumental-variables design of Miguel et al. (2004) using post-2005 weak-instrument diagnostics and an alternative panel identification strategy taught in our course. On the original panel of 41 sub-Saharan African countries (1981–1999), the rainfall first stage fails the Stock and Yogo (2005) 10%-maximum-size critical value by a factor of seven; the 2SLS point estimate carries the wrong sign and a cluster-bootstrap percentile interval that covers the entire interesting parameter space. A first-difference IV with the lagged growth level (Anderson and Hsiao, 1981), which does not depend on rainfall, recovers the negative MSS sign ($\hat{\beta}_{AH} = -0.35$) but with comparable imprecision. The data inform the sign of the income-conflict relationship but not its magnitude under either identifying assumption available within the original design.

1 Introduction

Estimating the causal effect of income shocks on civil conflict is not an estimation problem but an identification problem. OLS confounds three biases: conflict destroys output (reverse causation), national accounts in low-income economies are measured with error, and persistent country traits affect both (omitted variables). No amount of data fixes this. The effect is identified only if we can find a source of variation in growth that is exogenous to conflict and does not enter the conflict equation directly.

Miguel et al. (2004), henceforth MSS, proposed that source: annual rainfall growth, used as an excluded instrument for per-capita GDP growth on a panel of 41 sub-Saharan African countries over 1981–1999. The identifying assumptions are that rainfall shifts agricultural output (relevance), and that it affects conflict only through that channel (exclusion). The paper was novel in 2004. It also predates several of the diagnostic tools we now use to evaluate IV designs.

The literature that followed targeted each of these assumptions in turn. Brückner and Ciccone (2010) proposed an alternative instrument based on global commodity prices, relying on a different exogeneity argument. Ciccone (2011) questioned the specification of the first stage, showing that rainfall's mean-reversion contaminates the growth-as-IV interpretation. Sarsons (2015) tested the exclusion restriction directly, exploiting irrigation infrastructure that breaks the agricultural channel.

We revisit MSS with two questions. First, how much the original rainfall design tells us about β once weak-instrument diagnostics and a cluster bootstrap are applied to the standard 2SLS output. Second, whether panel-internal identification strategies that do not rely on rainfall, from Anderson and Hsiao (1981) first-difference IV to the difference- and system-GMM estimators of Arellano and Bond (1991) and Blundell and Bond (1998), recover a coherent estimate.

Section 2 formalises the MSS identification. Section 3 surveys the critiques. Section 4 describes the data. Section 5 sets out the estimators. Section 6 reports estimates. Section 7 concludes.

2 The MSS identification strategy

MSS fit the structural equation

$$C_{it} = \alpha_i + \lambda_i t + \beta\, g^y_{it} + \gamma' X_{it} + u_{it}, \tag{1}$$

where $C_{it} = \mathbf{1}\{\text{conflict}_{it}\}$ is a binary indicator for civil conflict in country i at year t, $g^y_{it} = \Delta\log(\text{GDP}_{it})$ is per-capita GDP growth, α_i are country fixed effects, $\lambda_i t$ are country-specific linear time trends, and X_{it} collects time-varying controls (lagged Polity 2). Opportunity-cost theory (Grossman, 1991) predicts $\beta < 0$.

Estimating equation (1) by OLS is inconsistent for the reasons listed in Section 1. MSS instrument g^y_{it} with annual rainfall growth g^R_{it} and its first lag:

$$g^y_{it} = a_i + b_i t + \pi_1\, g^R_{it} + \pi_2\, g^R_{i,t-1} + \delta' X_{it} + v_{it}. \tag{2}$$

The identifying assumptions, made conditional on α_i and X_{it}, are:

Relevance: $\text{Cov}(g^R_{it}, g^y_{it}) \neq 0$. Rainfall shifts agricultural output, which is a large share of African GDP. Testable through the first-stage F statistic on the excluded instruments.

Exclusion: $\text{Cov}(g^R_{it}, u_{it}) = 0$. Rainfall affects conflict only through GDP. Not directly testable in general; with two instruments, the Sargan test provides one degree of freedom of indirect check on whether the two rainfall measures imply the same β.

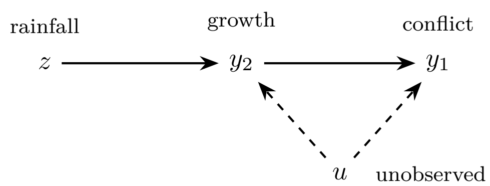

Figure 1: The MSS identification strategy as a DAG. Solid arrows show the assumed causal pathway: rainfall (z) affects growth (y_2), which affects conflict (y_1). Dashed arrows show unobserved confounders (u) that bias OLS. Identification of β requires the $z \to y_2$ arrow (relevance, testable) and the *absence* of any $z \to y_1$ arrow (exclusion, not directly testable).

The specification choices matter. MSS use country fixed effects to absorb time-invariant heterogeneity (geography, institutions, colonial history) and country-specific linear time trends to absorb slow secular movements (democratisation, technology diffusion). They do not use year fixed effects, which would absorb the common component of rainfall shocks across the panel and reduce the variation the instrument exploits. Standard errors are clustered at the country level.

Their preferred 2SLS specification yields $\hat{\beta} \approx -2.4$: a five-percentage-point decline in growth raises the probability of conflict by roughly twelve percentage points.

3 Methodological critiques and alternative identification

The literature that followed MSS extended the design along three axes: it offered alternative instruments, questioned the first-stage specification, and tested the exclusion restriction directly. Only the last two are critiques in the strict sense; the first is a parallel identification strategy.

Alternative instrument. Brückner and Ciccone (2010) replace rainfall with country-specific commodity export price shocks: the weighted sum of global commodity prices, using each country's export composition as weights. The exogeneity argument is small-country price-taking. Global commodity prices are determined on world markets, and no single sub-Saharan African country is large enough to move them. Their first stage becomes

$$g_{it}^{y} = a_i + \tilde{\pi}\,\mathrm{shock}_{it}^{P} + \tilde{\delta}' X_{it} + v_{it}, \tag{3}$$

while the structural equation (1) is unchanged. They recover a negative and significant β, corroborating the MSS sign under a different identifying assumption. This is identification pluralism rather than critique: two unrelated exogeneity arguments converge on the same conclusion, which strengthens MSS rather than weakens it. Within MSS's own design, the analogue is the Sargan over-identification test on the two rainfall instruments (current and lagged); we return to it in Section 6.

Specification critique. Ciccone (2011) attacks the functional form rather than the exogeneity. He shows that the original MSS result stems from a counterintuitive *positive* correlation between conflict at t and rainfall levels at $t-2$, rather than from negative rainfall shocks lowering income and raising conflict through the agricultural channel. Re-examining the analysis with updated data, he reports that conflict is largely unrelated to rainfall. Miguel and Satyanath (2011) reply showing that the MSS findings hold when rainfall levels are used as instruments, and that the first-stage relationship weakens after 2000. The lesson is generic: an exogenous instrument applied in the wrong functional form still produces an inconsistent estimate.

Direct test of exclusion. The most consequential critique tests the exclusion restriction itself. Sarsons (2015) uses Indian district-level data on rainfall and Hindu-Muslim religious riots, combined with records of irrigation dams. In districts *downstream of dams*, water supply is decoupled from local rainfall: income in these districts is much less sensitive to rainfall fluctuations, so the agricultural channel is mechanically attenuated. If MSS's exclusion holds (rainfall affects conflict only through agricultural income), then in dam-downstream districts rainfall should not predict conflict. She finds that rain shocks remain equally strong predictors of riot incidence in downstream districts. The implication is a direct $z \to y_1$ arrow that the MSS DAG denies (Figure 2).

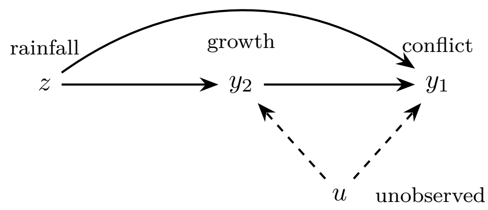

Figure 2: Sarsons's critique as a DAG. Same diagram as Figure 1, plus a direct $z \to y_1$ arrow that exclusion denies. Evidence from regions where the $z \to y_2$ channel is physically blocked (downstream of dams) supports the existence of this arrow.

The methodological lesson is general. Exclusion restrictions are sometimes partially testable: if the IV is claimed to operate through one channel, find a setting where the channel is shut and check whether the IV still moves the outcome. The post-2015 standard for IV papers in development economics has internalised this test.

4 Data

We use the replication panel distributed by MSS via the Harvard Dataverse (doi:10.7910/DVN/27324). It covers 41 sub-Saharan African countries over 1981–1999. After constructing one lag of GDP growth and dropping country-years with missing controls, the estimation sample is 702 country-year observations.

The outcome `any_prio` is a PRIO indicator equal to one if a country experiences at least 25 battle deaths in a year. The endogenous regressor `gdp_g` is annual per-capita GDP growth. The excluded instruments are current and lagged rainfall growth, `GPCP_g` and `GPCP_g_l`, constructed from the Global Precipitation Climatology Project. Controls follow MSS: initial log income, lagged Polity 2, ethnic and religious fractionalisation, an oil-exporter dummy, log mountainous terrain, and log lagged population. Only lagged Polity 2 is time-varying; the rest are absorbed by country fixed effects in our preferred specifications.

Table 1: Descriptive statistics, estimation sample ($N = 702$).

Variable	N	Mean	SD	Min	Max
`any_prio`	702	0.269	0.444	0.000	1.000
`gdp_g`	702	−0.006	0.066	−0.474	0.319
`GPCP_g`	702	0.015	0.210	−0.550	1.680
`y_0`	702	1.16	0.90	0.316	4.84
`polity2l`	702	−3.55	5.55	−10	9
`ethfrac`	702	0.655	0.237	0.036	0.925
`relfrac`	702	0.487	0.185	0.000	0.783
`Oil`	702	0.117	0.321	0	1
`lmtnest`	702	1.58	1.43	0	4.42
`lpopl1`	702	8.77	1.20	5.75	11.7

Three features of the panel matter for what follows. The unconditional conflict frequency is 27%. Average growth is slightly negative (-0.6% per annum). The panel is democracy-poor on average (mean Polity 2 of -3.5 on a scale of -10 to 10).

5 Methodology

Each estimator below corresponds to a different way of reading the DAG in Figure 1. POLS uses all variation jointly and trusts that the dashed $u \to y_2$ and $u \to y_1$ arrows are weak. FE removes the time-invariant part of u. 2SLS exploits the $z \to y_2$ arrow and assumes no direct $z \to y_1$ arrow. The cluster bootstrap provides an alternative inference for $\hat{\beta}_{\text{2SLS}}$ when conventional asymptotics are unreliable. First-difference IV with lagged levels identifies β without using z at all, relying on internal panel variation instead.

5.1 Pooled OLS (LPM)

We begin by estimating equation (1) by OLS, treating the binary outcome as a linear probability model. In subscript form,

$$\hat{\beta}_{\text{POLS}} = \left(\textstyle\sum_{it} x_{it}x_{it}'\right)^{-1} \textstyle\sum_{it} x_{it}C_{it} \quad \Longleftrightarrow \quad \hat{\beta}_{\text{POLS}} = (X'X)^{-1}X'C,$$

where the right-hand expression stacks all NT observations into matrices X and C. Consistency requires $\text{Cov}(g^y_{it}, u_{it}) = 0$, which the three endogeneity sources in Section 1 rule out. POLS therefore serves as a benchmark, not as a causal estimate. Two issues are handled at the inference stage. First, the LPM is heteroscedastic by construction since $\text{Var}(u \mid x) = p(x)(1 - p(x))$ depends on x; we report the Breusch-Pagan and White tests as confirmation. Second, errors are correlated within country across years; we cluster standard errors at the country level throughout.

5.2 Fixed and random effects

Fixed-effects estimation removes the time-invariant part of u. Let $u_{it} = c_i + e_{it}$ with c_i constant within country. The within transformation subtracts country means from each variable: let $\tilde{y}_{it} = y_{it} - \bar{y}_i$ and $\tilde{x}_{it} = x_{it} - \bar{x}_i$. The resulting estimator is

$$\hat{\beta}_{\text{FE}} = \left(\textstyle\sum_{it} \tilde{x}_{it}\tilde{x}_{it}'\right)^{-1} \textstyle\sum_{it} \tilde{x}_{it}\tilde{y}_{it},$$

which is consistent if $\text{Cov}(g^y_{it}, e_{it}) = 0$ after demeaning. We report two-way FE (country and year) as our baseline and add a specification with country-specific linear trends in place of year FE, the original MSS choice. The two differ in how they treat the common component of rainfall variation across the panel: year FE absorb it, country trends preserve it.

Random effects is consistent under the stronger assumption $\text{Cov}(c_i, x_{it}) = 0$ and is more efficient when it holds. The Hausman test compares FE and RE: a large χ^2 statistic indicates the RE assumption fails. Time-invariant controls (initial income, fractionalisation, terrain, oil, lagged population) are absorbed by country FE and do not appear in the FE or RE specifications.

5.3 Two-stage least squares

We then implement the MSS strategy: instrument g^y_{it} with current and lagged rainfall growth from equation (2). As a two-stage procedure, run the first stage in equation (2) by OLS to obtain fitted values $\hat{g}^y_{it}$, then plug those into the structural equation:

$$C_{it} = \alpha_i + \lambda_i t + \beta\, \hat{g}^y_{it} + \gamma' X_{it} + \text{error}_{it}.$$

In matrix form this is equivalent to

$$\hat{\beta}_{\text{2SLS}} = (X' P_Z X)^{-1} X' P_Z C,$$

where $P_Z = Z(Z'Z)^{-1}Z'$ projects onto the column space of the instruments (country FE and exogenous controls included in Z). Two assumptions support consistency.

Relevance. We test it through the cluster-robust F statistic on the excluded instruments in the first stage (Staiger and Stock, 1997), compared to the Stock and Yogo (2005) 10%-maximum-size critical value of 19.93 (two IVs, one endogenous regressor). Values below the threshold indicate weak instruments and invalid conventional standard errors.

Exclusion. Not directly testable. With two instruments we report the Sargan over-identification test, which asks whether both rainfall measures imply the same β. Rejection indicates that at least one instrument is correlated with u. Failure to reject is weak evidence at best given the single degree of freedom and the fact that both instruments share the same exogeneity argument; the test cannot detect homogeneous invalidity.

We additionally report the Wu-Hausman endogeneity test, which compares OLS and 2SLS and rejects exogeneity of g^y_{it} if the two estimates differ by more than sampling noise.

5.4 Cluster bootstrap inference

When the first-stage F falls short of conventional critical values, the standard 2SLS t-statistic does not have an asymptotic standard normal distribution and reported confidence intervals can have actual coverage far below their nominal level. We supplement the conventional output with a non-parametric cluster bootstrap. Countries are resampled with replacement, the 2SLS coefficient is re-estimated on each resampled panel, and the percentile interval is reported from $B = 500$ replications. Resampling at the country level preserves the within-country correlation structure that the cluster-robust standard errors of Section 5.1 assume.

5.5 First-difference IV with lagged levels

If rainfall is too weak to identify β externally, the panel itself contains a second source of variation. First-differencing equation (1) eliminates the country fixed effects:

$$\Delta C_{it} = \beta\, \Delta g^y_{it} + \gamma' \Delta X_{it} + \Delta u_{it}.$$

Anderson and Hsiao (1981) propose instrumenting the differenced regressor Δg^y_{it} with its lagged level $g^y_{i,t-1}$, under the assumption $\text{Cov}(g^y_{i,t-1}, \Delta u_{it}) = 0$: past growth is uncorrelated with the change in unobserved conflict shocks. This identifying assumption is logically separate from the rainfall exclusion restriction; the two strategies fail or succeed for different reasons.

5.6 Difference and system GMM

Conflict is highly persistent: the lagged outcome is a strong predictor of current conflict. Extending equation (1) with $C_{i,t-1}$ as a regressor turns it into a dynamic panel,

$$C_{it} = \alpha_i + \rho\, C_{i,t-1} + \beta\, g^y_{it} + \gamma' X_{it} + u_{it}. \tag{4}$$

Fixed effects on a dynamic panel is inconsistent for finite T (Nickell, 1981). Arellano and Bond (1991) first-difference equation (4) to eliminate α_i and instrument the differenced lagged outcome $\Delta C_{i,t-1}$ with deeper lags of the level outcome $C_{i,t-2}, C_{i,t-3}, \ldots$; the endogenous Δg^y_{it} is instrumented similarly. Blundell and Bond (1998) extend this with additional moment conditions on the levels equation (system-GMM), more efficient when the series are persistent. We use two-step estimators with cluster-robust SE and collapse the instrument matrix to lags two and three to keep the instrument count well below $N = 41$ (Roodman, 2009). Validity is assessed via the Sargan-Hansen test on the over-identifying restrictions and the Arellano-Bond m_2 test for second-order autocorrelation in the differenced residuals; failure to reject m_2 is a necessary condition for the moment conditions to hold. These estimators are formally derived for continuous outcomes; we treat the linear probability specification of C_{it} as a defensible approximation that is common in the conflict literature.

6 Results and diagnostics

Table 2 reports the seven static specifications described in Section 5. Tables 3 and **??** report the diagnostic battery and the dynamic-panel GMM estimates.

Table 2: Static specifications. Cluster-robust SE at country level in parentheses. Models 2–6 use two-way FE; model 7 replaces year FE with country-specific linear time trends. Time-invariant controls are included in POLS only and absorbed by country FE elsewhere.

	POLS (LPM)	FE 2-way	RE 2-way	Red. form	First stage	2SLS	2SLS (trends)
GDP growth	-0.507^{**}	-0.460^{**}	-0.443^{**}			0.141	-1.297
	(0.222)	(0.220)	(0.217)			(1.330)	(1.071)
GDP growth (lag)	0.020	-0.173	-0.121			-0.165	0.040
	(0.243)	(0.221)	(0.277)			(0.229)	(0.185)
Rainfall growth				0.024	0.042^{**}		
				(0.055)	(0.018)		
Rainfall growth (lag)				-0.084^{*}	0.025		
				(0.048)	(0.016)		
Polity 2 (lag)	0.000	-0.004	-0.002	-0.003	-0.001^{*}	-0.003	-0.004
	(0.004)	(0.005)	(0.005)	(0.005)	(0.001)	(0.006)	(0.005)
N	702	702	702	702	702	702	702
R^2	0.129	0.012	0.006	0.005	0.019	0.569	0.710

$^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

POLS reproduces the MSS direction: a one-standard-deviation drop in growth (≈ 6.6 percentage points) raises the conflict probability by $0.51 \times 0.066 \approx 3.4$ percentage points, significant at the 5% level. FE and RE shrink the coefficient slightly towards zero ($\hat{\beta} = -0.46, -0.44$) with widened standard errors, as expected when between-country variation is stripped out. The reduced form shows that rainfall growth enters the conflict equation only weakly (lagged rainfall has $p = 0.08$). The first stage confirms that current-year rainfall growth has a statistically significant but quantitatively modest effect on GDP growth.

The 2SLS estimate (column 6) is the striking column. The point estimate is +0.14 with a cluster-robust SE of 1.33: the wrong sign, no statistical significance, and a confidence interval covering essentially all economically relevant magnitudes. This collapse of the IV estimate relative to the well-determined POLS estimate is the canonical fingerprint of a weak instrument. Column 7 re-estimates the same model with country FE plus country-specific linear time trends (the MSS specification) and recovers $\hat{\beta} = -1.30$ (SE 1.07, $p = 0.23$): the correct sign, larger in magnitude than OLS, but still imprecisely estimated. The qualitative direction of MSS is therefore sensitive to whether secular country-level trends are modelled.

Table 3: Specification and identification diagnostics.

Statistic	Value	p-value
Heteroscedasticity		
Breusch–Pagan	142.52	$< 10^{-16}$
Panel structure		
F-test country FE ($H_0\colon c_i = 0$)	14.90	$< 10^{-16}$
Hausman FE vs RE	0.04	0.998
Weak instruments		
First-stage F baseline (cluster-robust)	2.89	0.056
First-stage F MSS-trends (cluster Wald)	4.45	0.012
First-stage F MSS-trends (homoscedastic)	7.48	0.0006
Stock–Yogo 10%-maximum-size critical value	19.93	—
Endogeneity		
Wu–Hausman	0.14	0.705
Overidentification		
Sargan	2.40	0.121
Bootstrap inference		
Cluster-bootstrap 95% CI for β^{IV}	[−3.52, 3.51]	—

The diagnostics tell a coherent story. Heteroscedasticity is decisively present, confirming the necessity of robust standard errors for the LPM. Country fixed effects are jointly significant, but the Hausman test returns a vanishing statistic that we discount: the test's construction assumes the FE estimator is efficient under H_0, which fails once standard errors are clustered, and its power is limited in moderately unbalanced panels. We default to FE.

The weak-instrument diagnostics are the crux. The first-stage F on the two excluded rainfall instruments is 2.89 in our baseline, roughly one-seventh of the Stock–Yogo threshold of 19.93. The MSS-trends specification raises it to 4.45 (cluster-robust) or 7.48 (homoscedastic), notably stronger but still below the threshold. Conventional 2SLS inference is therefore unreliable in both specifications, and the Wu–Hausman test fails to reject exogeneity ($p = 0.705$): in this sample we have no positive statistical evidence that OLS is biased, though theory says it must be.

A non-parametric cluster bootstrap (500 replications, resampling countries with replacement) yields a 95% percentile interval of $[-3.52, 3.51]$, wide enough to include zero and most economically relevant magnitudes. The Sargan test on the two rainfall instruments fails to reject ($p = 0.121$), which we read as weak reassurance only: with one degree of freedom and two instruments that share their exogeneity argument, the test cannot detect homogeneous invalidity of the kind Sarsons (2015) flags.

Table 4: Panel-internal identification strategies. Column 1 is the Anderson and Hsiao (1981) FD-IV on the static structural equation; columns 2 and 3 are two-step difference- and system-GMM on the dynamic specification (4), with collapsed instrument matrices at lags 2–3. Cluster-robust SE at country level in parentheses.

	Anderson-Hsiao	Arellano-Bond	Blundell-Bond
Conflict (lag)		0.334^{**}	0.628^{***}
		(0.142)	(0.095)
GDP growth	-0.349	-0.293^{*}	-0.296^{*}
	(0.383)	(0.150)	(0.155)
GDP growth (lag)		0.044	-0.046
		(0.211)	(0.225)
Polity 2 (lag)	-0.005	0.000	-0.006^{*}
	(0.005)	(0.004)	(0.003)
N (country-years)	620	—	—
First-stage F on $g^y_{i,t-1}$	451.7	—	—
Sargan-Hansen p	—	0.51	0.09
m_2 (AR(2)) p	—	0.50	0.47

$^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The three estimators tell a consistent qualitative story under identifying assumptions that differ. Anderson-Hsiao gives $\hat{\beta}_{AH} = -0.349$ with cluster-robust SE 0.383. The first-stage F on the lagged level $g^y_{i,t-1}$ is 451.7, far above the Stock-Yogo 10%-maximum-size critical value of 16.4 for one instrument and one endogenous regressor, but the strength reflects mean reversion in the country-level growth series rather than new identifying variation. Identification rests on $\mathrm{Cov}(g^y_{i,t-1}, \Delta u_{it}) = 0$, which is independent from the rainfall exclusion.

The Arellano-Bond difference-GMM estimator, which adds $C_{i,t-1}$ to the equation and uses deeper lags as instruments, returns $\hat{\beta}_{AB} = -0.293$ ($p = 0.05$). Blundell-Bond yields a similar point estimate -0.296 but the standard error widens slightly and significance drops below the 10% level. The lagged conflict coefficient is large and highly significant in system-GMM ($\hat{\rho} = 0.628$, $p < 0.001$), confirming substantial persistence. The Sargan-Hansen tests do not reject for either estimator ($p = 0.51$ for AB, $p = 0.09$ for BB, the latter borderline); the m_2 tests do not detect second-order autocorrelation in differences ($p = 0.50$ and 0.47). Both validations support the identifying restrictions.

Across all three panel-internal designs the MSS sign reappears, though the magnitude is pinned down only loosely. None of these strategies uses rainfall.

7 Discussion

The empirical findings are mixed and the appropriate reading depends on what one is willing to assume about identification.

OLS recovers the qualitative MSS direction with a tight standard error: $\hat{\beta}_{\text{POLS}} = -0.507$. Under exogeneity of growth (which the three endogeneity sources of Section 1 rule out), this would be the most precise summary of the conditional relationship between growth and conflict. As a causal estimate, it is not defensible.

Under the MSS rainfall identification, conventional inference is unreliable. The first-stage F of 2.89 in our two-way FE specification is roughly one-seventh of the Stock-Yogo threshold; the 2SLS point estimate carries the wrong sign and a confidence interval that covers the entire interesting parameter space. Restoring the MSS country-trend specification raises the first stage to 4.45 and recovers $\hat{\beta} = -1.30$, of the correct sign but still imprecisely estimated. The cluster bootstrap percentile interval $[-3.52, 3.51]$ confirms the order of magnitude of the irreducible uncertainty in this design under conventional 2SLS, though the bootstrap addresses finite-sample clustered inference and does not itself restore identification under weak instruments.

Under the panel-internal identification strategies of Section 5.5 and 5.6, the negative sign reappears across all three estimators ($\hat{\beta}_{\text{AH}} = -0.349$, $\hat{\beta}_{\text{AB}} = -0.293$, $\hat{\beta}_{\text{BB}} = -0.296$), with Arellano-Bond returning a coefficient significant at the 10% level. The Anderson-Hsiao first-stage strength is mechanical, driven by mean reversion in growth, but the GMM estimators rest on deeper lag-orthogonality and (for system-GMM) initial-condition restrictions that are logically distinct from the rainfall exclusion.

What the data support: the qualitative sign of the MSS effect is robust across credible specifications under at least two distinct identifying assumptions (rainfall exclusion, past-growth orthogonality). What the data do not support: a precise magnitude, or a credible defence of either identifying assumption against the Sarsons (2015) critique, which targets the rainfall channel directly. The Sargan over-identification test on the two rainfall instruments does not reject, but with one degree of freedom and two instruments that share their exogeneity argument the test has neither the power nor the structure to detect the kind of violation that matters.

The most honest reading is that the data inform the sign of β but not its magnitude. A precise causal estimate of how much income shocks raise the probability of civil conflict in this sample requires either a stronger instrument, the commodity-price approach of Brückner and Ciccone (2010) being the leading candidate, or a research design that does not rely on rainfall at all.

References

Anderson, T.W., and C. Hsiao (1981). "Estimation of dynamic models with error components." *Journal of the American Statistical Association* 76(375): 598–606.

Arellano, M., and S. Bond (1991). "Some tests of specification for panel data: Monte Carlo evidence and an application to employment equations." *Review of Economic Studies* 58(2): 277–297.

Blundell, R., and S. Bond (1998). “Initial conditions and moment restrictions in dynamic panel data models.” *Journal of Econometrics* 87(1): 115–143.

Brückner, M., and A. Ciccone (2010). “International commodity prices, growth and the outbreak of civil war in Sub-Saharan Africa.” *Economic Journal* 120(544): 519–534.

Ciccone, A. (2011). “Economic shocks and civil conflict: A comment.” *American Economic Journal: Applied Economics* 3(4): 215–227.

Grossman, H.I. (1991). “A general equilibrium model of insurrections.” *American Economic Review* 81(4): 912–921.

Miguel, E., S. Satyanath, and E. Sergenti (2004). “Economic shocks and civil conflict: An instrumental variables approach.” *Journal of Political Economy* 112(4): 725–753.

Miguel, E., and S. Satyanath (2011). “Re-examining economic shocks and civil conflict.” *American Economic Journal: Applied Economics* 3(4): 228–232.

Nickell, S. (1981). “Biases in dynamic models with fixed effects.” *Econometrica* 49(6): 1417–1426.

Roodman, D. (2009). “A note on the theme of too many instruments.” *Oxford Bulletin of Economics and Statistics* 71(1): 135–158.

Sarsons, H. (2015). “Rainfall and conflict: A cautionary tale.” *Journal of Development Economics* 115: 62–72.

Staiger, D., and J.H. Stock (1997). “Instrumental variables regression with weak instruments.” *Econometrica* 65(3): 557–586.

Stock, J.H., and M. Yogo (2005). “Testing for weak instruments in linear IV regression.” In: *Identification and Inference for Econometric Models: Essays in Honor of Thomas Rothenberg*, ed. D.W.K. Andrews and J.H. Stock, Cambridge University Press, 80–108.